Dietary acrylamide intake and the risk of renal cell, bladder, and prostate cancer.

. . . . .

BACKGROUND: Acrylamide, a probable human carcinogen, was recently detected in various heat-treated carbohydrate-rich foods. Epidemiologic studies on the relation with cancer have been few and largely negative. OBJECTIVE: We aimed to prospectively examine the association between dietary acrylamide intake and renal cell, bladder, and prostate cancers. DESIGN: The Netherlands Cohort Study on diet and cancer includes 120,852 men and women aged 55-69 y. At baseline (1986), a random subcohort of 5000 participants was selected for a case-cohort analysis approach using Cox proportional hazards analysis. Acrylamide intake was assessed with a food-frequency questionnaire at baseline and was based on chemical analysis of all relevant Dutch foods. RESULTS: After 13.3 y of follow-up, 339, 1210, and 2246 cases of renal cell, bladder, and prostate cancer, respectively, were available for analysis. Compared with the lowest quintile of acrylamide intake (mean intake: 9.5 microg/d), multivariable-adjusted hazard rates for renal cell, bladder, and prostate cancer in the highest quintile (mean intake: 40.8 microg/d) were 1.59 (95% CI: 1.09, 2.30; P for trend = 0.04), 0.91 (95% CI: 0.73, 1.15; P for trend = 0.60), and 1.06 (95% CI: 0.87, 1.30; P for trend = 0.69), respectively. There was an inverse nonsignificant trend for advanced prostate cancer in never smokers. CONCLUSIONS: We found some indications for a positive association between dietary acrylamide and renal cell cancer risk. There were no positive associations with bladder and prostate cancer risk.